

We finish this section observing that in 1D we can say something more.

Proposition 7.30. *Suppose $d = 1$, $\Omega = [a, b] \subset \mathbb{R}$ and take $q > 0$. Then the functional F defined through $F(\rho) = \int_a^b \rho(x)^{-q} dx$ for $\rho \ll \mathcal{L}^1$ is displacement convex in $\mathbb{W}_p(\Omega)$. In particular, if $\rho_0, \rho_1 \geq c > 0$, then we also have $\rho_t(x) \geq c$ for every $t \in [0, 1]$, where ρ_t is the geodesic connecting ρ_0 to ρ_1 .*

Proof. Notice the the function $f(s) = s^{-q}$ is convex for $s > 0$ and that $s \mapsto s^d f(s^{-d}) = s^{q+1}$ is convex but increasing. Yet, if we re-read the proof of Theorem 7.28 we observe that we only needed it to be decreasing to ensure that the composition with g was convex, but in this case g is affine, and convexity of $s \mapsto s^d f(s^{-d})$ is enough. Moreover, if we assume $F(\rho) < +\infty$ then we have $\rho > 0$ a.e., which shows that it is not necessary to assume $f(0) = 0$. This proves the desired displacement convexity. To get the lower bound on geodesics, we just need to act as in proposition 7.29, letting $q \rightarrow \infty$. $\square$ $\square$

7.3.3 Convexity on generalized geodesics

It is quite disappointing to note that the functional $\mu \mapsto W_2^2(\mu, \nu)$ is not, in general, displacement convex. This seems contrary to the intuition because usually squared distances are nice convex functions. However, we can see that this fails from the following easy example. Take $\nu = \frac{1}{2}\delta_{(1,0)} + \frac{1}{2}\delta_{(-1,0)}$ and $\mu_t = \frac{1}{2}\delta_{(t,a)} + \frac{1}{2}\delta_{(-t,-a)}$. The curve μ_t , if $a > 1$ is the geodesic between μ_{-1} and μ_1 (because the optimal transport between these measures sends $(a, -1)$ to $(a, 1)$ and $(-a, 1)$ to $(-a, -1)$). Yet, if we compute $W_2^2(\mu_t, \nu)$ we have

$$W_2^2(\mu_t, \nu) = a^2 + (1-t)^2 \wedge (1+t)^2.$$

But this function is not convex! (see Figure 7.1)

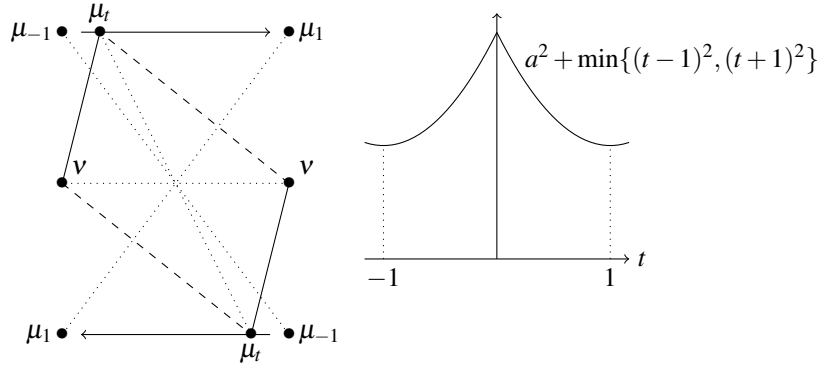


Figure 7.1: The distance $W_2^2(\mu_t, \nu)$.

The lack of geodesic convexity of this easy functional⁴ is a problem for many issues (in particular for some metric approaches to gradient flows through the EVI condition,

⁴By the way, this functional can even be prove to be somehow geodetically concave, as it is shown in [17], Theorem 7.3.2.

see Section 8.4.1) and an alternate notion has been proposed, namely that of convexity along generalized geodesics.

Definition 7.31. If we fix an absolutely continuous probability $\rho \in \mathcal{P}(\Omega)$, for every pair $\mu_0, \mu_1 \in \mathcal{P}(\Omega)$ we call *generalized geodesic* between μ_0 and μ_1 with base ρ in $\mathbb{W}_2(\Omega)$ the curve $\mu_t = ((1-t)T_0 + tT_1)_\# \rho$, where T_0 is the optimal transport map (for the cost $|x-y|^2$) from ρ to μ_0 , and T_1 from ρ to μ_1 .

It is clear that $t \mapsto W_2^2(\mu_t, \rho)$ satisfies

$$\begin{aligned} W_2^2(\mu_t, \rho) &\leq \int |((1-t)T_0(x) + tT_1(x)) - x|^2 d\rho(x) \\ &\leq (1-t) \int |T_0(x) - x|^2 d\rho(x) + t \int |T_1(x) - x|^2 d\rho(x) \\ &= (1-t)W_2^2(\mu_0, \rho) + tW_2^2(\mu_1, \rho) \end{aligned}$$

and hence we have the desired convexity along this curve. Moreover, similar considerations to those we developed in this section show that all the functionals that we proved to be geodesically convex are also convex along generalized geodesics. We do not develop these proofs here, and we refer to [17] for more details (for the case of the functional $\mathcal{F}$, Lemma 7.26 has to be changed into “ $t \mapsto \det((1-t)A + tB)^{1/d}$ is concave, whenever A and B are symmetric and positive-definite” (the proof is similar).

7.4 Discussion

7.4.1 A case study: $\min F(\rho) + W_2^2(\rho, \nu)$

Among variational problems involving optimal transportation and Wasserstein distances, a very recurrent one is the following

$$\min_{\rho \in \mathcal{P}_2(\Omega)} \frac{1}{2} W_2^2(\rho, \nu) + \tau F(\rho), \quad (7.2)$$

where F is a given functional on probability measures, $\tau > 0$ a parameter which can possibly be small, and ν is a given probability in $\mathcal{P}_2(\Omega)$ (the space of probability measures on Ω with finite second moment $\int |x|^2 d\rho(x) < +\infty$). This very instance of the problem is exactly the one we face in the time-discretization of the gradient flow of F in $\mathbb{W}_2(\Omega)$ (see Section 8.2).

But the same problem also appears, for fixed $\tau > 0$, in other frameworks as well. For instance in image processing, if F is a smoothing functional, this is a model to find a better (smoother) image ρ which is not so far from the original density ν (and the choice of the distance W_2 is justified by robustness arguments), see [209, 93]. In urban planning (see [94, 270]) ν can represent the distribution of some resources and ρ that of population, who wants to be close to ν but also to guarantee enough space to each individual. In this case the functional F favors diffuse measures (for instance using the functional $\mathcal{F}(\rho) = \int f(\rho(x)) dx$ for a convex and superlinear function f

with $f(0) = 0$, which gives a higher cost to high densities of ρ and is minimized, on bounded domains, by the uniform measure). Reciprocally, ν could instead represent the distribution of population, and ρ that of services, to be chosen so that they are close enough to ν but more concentrated. In this case F will favor concentrated measures. When F penalizes the number of points in the support of ρ , then ρ becomes finitely atomic and we face the so-called location problem (finding where to place a finite number of points so as to approximate a given continuous distribution), useful both in logistics and in quantization issues in signal processing. See for instance [74, 76] for a transport-based analysis of the asymptotics of this problem.

Note that, when F only takes value 0 and $+\infty$, the above problem becomes a projection problem. In particular (see Section 8.4.2), the projection onto the set of densities bounded above by the constant 1 has received lot of attention because of its applications in the time-discretization of evolution problems with density constraints, in particular for crowd motion (see [265, 224]), where a crowd is described as a population of particles which cannot overlap, and cannot go beyond a certain threshold density).

In this section we would like to briefly discuss the characterization and regularity of the optimizer, at least in the case where $\mathcal{F}(\rho) = \int f(\rho)$ for a nice convex integrand $f : \mathbb{R}_+ \rightarrow \mathbb{R}$, superlinear at infinity (i.e. one of the main functionals studied in this chapter).

We are interested in the estimates that one can give on the minimizer $\hat{\rho}$. They can be roughly divided into two categories: those which are independent of ν but depend on τ , and those which are uniform in τ but refer to similar bounds on ν .

For instance, let us write down the optimality conditions for (7.2) in the case $\mathcal{F}(\rho) = \int f(\rho)$, using Proposition 7.20. First we suppose that $f \in C^2$, that we have $0 < c \leq f'' \leq C$, and that $\text{spt}(\nu) = \Omega$. This allows to differentiate both terms in the functional (the assumption on ν is needed to have uniqueness of the c -concave Kantorovich potential) and we get

$$\begin{cases} \varphi + \tau f'(\hat{\rho}) = C & \hat{\rho} - \text{a.e.}, \\ \varphi + \tau f'(\hat{\rho}) \geq C & \text{on } \{\hat{\rho} = 0\}, \end{cases} \quad (7.3)$$

where φ is a suitable Kantorovich potential in the transport from $\hat{\rho}$ to ν . We can write the above condition as

$$\hat{\rho} = (\tau f')^{-1}((C - \varphi) \vee f'(0)). \quad (7.4)$$

This provides a uniform Lipschitz bound on ρ , which only depends on the Lipschitz constant of φ (which depends on $\text{diam}(\Omega)$) and of $(f')^{-1}$ (which is bounded by c^{-1}). In particular, if we let f and ν vary, keeping the condition $f'' \geq c > 0$, we can apply Ascoli-Arzelà Theorem to ρ and pass to the limit (from the uniqueness of the minimizers, which is guaranteed by the strict convexity of f , the minimizers pass to the limit). Hence, we can guarantee that (7.4) holds under the only assumption $f'' \geq c > 0$. Notice that in the case $f'(0) = -\infty$ we get $\hat{\rho} = (\tau f')^{-1}(C - \varphi)$, that we prefer to re-write as $\tau f'(\hat{\rho}) = C - \varphi$.

These considerations allow to get continuity for $\hat{\rho}$, but the bounds obviously degenerate as $\tau \rightarrow 0$. On the other hand, they do not really depend on ν .

A different type of bound that one can prove is the following one

Proposition 7.32. *If $g \in L^\infty$ is a given probability density on a compact convex and smooth domain Ω , and $\hat{\rho}$ solves*

$$\min_{\rho \in \mathcal{P}(\Omega)} \frac{1}{2} W_2^2(\rho, g) + \int f(\rho(x)) dx,$$

where f is convex and superlinear, then $\|\hat{\rho}\|_{L^\infty} \leq \|g\|_{L^\infty}$

This bound, which, on the contrary, is independent of τ , is proven in [111, 270]. We can provide a sketch of proof.

Proof. We start from the case where g is smooth and strictly positive, and f is smooth on $]0, +\infty[$, continuous on $\mathbb{R}_+$, with $f'' \geq c > 0$ and $f'(0) = -\infty$. From the above considerations (with $\tau = 1$), we know that the density of the optimal $\hat{\rho}$ is a continuous function, given by $\hat{\rho} = (f')^{-1}(C - \varphi)$. Note that $(f')^{-1}$ is Lipschitz continuous, as a consequence of the lower bound on f'' , and that $\hat{\rho} > 0$ since $\hat{\rho} = 0$ would imply $-\varphi = -\infty$. From its continuity, $\hat{\rho}$ admits a maximal value, $\hat{\rho}(x_0)$. Note that a maximum point for $\hat{\rho}$ corresponds to a minimum point for φ . Using Caffarelli's regularity theory (see Section 1.7.6) we can provide enough regularity for φ so as to justify all the following considerations (indeed, since g is smooth and $\hat{\rho}$ is Lipschitz, and both are strictly positive, and hence bounded from below, on a convex set, one has at least $\varphi \in C^{2,\alpha}$). First, suppose that $x_0 \notin \partial\Omega$. In this case we should have $D^2\varphi(x_0) \geq 0$. But the Monge-Ampère equation gives us

$$\hat{\rho}(x_0) = \det(I - D^2\varphi(x_0))g(T(x_0)),$$

where $T = \text{id} - \nabla\varphi$ is the optimal transport from $\hat{\rho}$ to g . From $D^2\varphi(x_0) \geq 0$ we get $\det(I - D^2\varphi(x_0)) \leq 1$, and hence

$$\|\hat{\rho}\|_{L^\infty} = \hat{\rho}(x_0) \leq g(T(x_0)) \leq \|g\|_{L^\infty}.$$

It remains to consider the case $x_0 \in \partial\Omega$. Since $T(x_0) = x_0 - \nabla\varphi(x_0) \in \overline{\Omega}$ and Ω is convex, we have $\nabla\varphi(x_0) \cdot \mathbf{n} \geq 0$. By the optimality property of x_0 (which is a minimum point for φ), we also have the opposite inequality, and hence we obtain anyway $\nabla\varphi(x_0) \cdot \mathbf{n} = 0$. This allows to conclude that also in this case we have $D^2\varphi(x_0) \geq 0$ (indeed, second order minimality conditions are also satisfied on the boundary if we already know that the gradient vanish; since the tangential part always vanish at extremal points, only the normal one has to be considered).

Finally, one can get rid of the assumptions on g and f by approximation (since we know that the minimizer of the original problem with $g \in L^\infty$ and f which is only convex and superlinear is unique). $\square$ $\square$

As a final remark, we stress that some limited higher-order similar estimates can also be obtained, and are the object of very recent papers. We mention a BV estimate, namely $\|\hat{\rho}\|_{BV} \leq \|g\|_{BV}$, presented in [152], and based on some integration-by-parts techniques. We also mention a Lipschitz estimate which, together with some semi-concavity bounds, is proved in [208] in the case where $f(t) = t \log t$, thanks to some Pogorelov-type methods. Both estimates can be adapted to the case where a potential energy $\mathcal{V}$ is added to the functional. However, we do not develop details here.

7.4.2 Brunn-Minkowski inequality

Another interesting consequence of the displacement convexity in $\mathbb{W}_2$ is a transport-based proof of a well-known geometric inequality called Brunn-Minkowski inequality. This inequality states that for every two sets $E, F \subset \mathbb{R}^d$ (we ignore here measurability issues) we have

$$|E + F|^{1/d} \geq |E|^{1/d} + |F|^{1/d}, \text{ where } E + F = \{x + y : x \in E, y \in F\}.$$

This inequality can be proven in the following way. Consider the measures μ with constant density $\frac{1}{|E|}$ on E and ν with constant density $\frac{1}{|F|}$ on F . Then, take the measure $\mu_{1/2}$ obtained at time $t = 1/2$ on the constant speed geodesic between them. Since $\mu_{1/2} = (\frac{1}{2}\text{id} + \frac{1}{2}T)_{\#}\mu_0$ we know that $\mu_{1/2}$ is concentrated on $A := \frac{1}{2}E + \frac{1}{2}F$. Moreover, $\mu_{1/2}$ is absolutely continuous and we call ρ its density (we know $\rho \in L^\infty$ because $\mu, \nu \in L^\infty$, thanks to Proposition 7.29). Note that, for convex f , we have

$$\frac{1}{|A|} \int_A f(\rho(x)) dx \geq f\left(\frac{\int_A \rho(x) dx}{|A|}\right) = f(|A|^{-1}).$$

Geodesic convexity of $\mathcal{F}$ for $f(t) = -t^{1-1/d}$ implies

$$\begin{aligned} -\frac{1}{2}|E|^{1/d} - \frac{1}{2}|F|^{1/d} &= \frac{1}{2}\mathcal{F}(\mu) + \frac{1}{2}\mathcal{F}(\nu) \geq \mathcal{F}(\mu_{1/2}) \\ &= \int_A f(\rho(x)) dx \geq |A|f(|A|^{-1}) = -|A|^{1/d}. \end{aligned}$$

If we multiply this inequality by -2 on each side, using the scaling properties of the Lebesgue measure, we exactly obtain the Brunn-Minkowski inequality.

7.4.3 Displacement convexity, game theory and spatial economics: urban equilibria

As optimal transport describes the cost of moving masses (or individuals) when their density is known, it is a very natural tool to analyze problems in spatial economics, where densities of agents and of activities are involved, and one wants to take into account geographical effects. Often, it is important to associate to this densities some “performances”, which let the functionals that we analyzed in this chapter appear. Among the objects studied in spatial economics, one of the most intriguing, as we can see in our everyday life, is urban structure.

Many models exist in the economic literature and most of them have nothing to do with optimal transport. An interesting treatise of some spatial problems involving some notions of optimal transform (more under the form of flow-minimization, such as in Chapter 4, then under the form of a Monge-Kantorovich problem) is contained in the book [29], which inspired by the way some of the works on congested transport contained in Section 4.4.1.

Yet, the goal of this section is not to give a comprehensive literature review on the subject, but to point out some recent problems in urban economics where the analysis

we did in this chapter is particularly relevant. Indeed, we will see that displacement convexity can play a role in proving uniqueness of equilibria, and their characterization as the minimizers of a global energy.

In particular, we will not spend words on the models for urban planning contained in [94, 95], where the goal is to find the optimal shapes of the distribution μ of the population and of the distribution ν of “services” in order to minimize a global quantity $\mathcal{F}(\mu) + \mathcal{T}_c(\mu, \nu) + \mathcal{G}(\nu)$. They are problems involving a social planner where no equilibrium issue is addressed, and optimal transport appears directly in the modeling and not as a tool. We will not spend word neither on the equilibrium model contained in [107, 108], where one looks at a distribution μ of residents, a distribution ν of jobs, a pairing γ between them, a function φ giving the price of the land and a function ψ giving wages and at other related quantities. Here an equilibrium may be found by considering φ and ψ as Kantorovich potentials for an appropriate cost, but the model is quite long to describe (see also [277] for a short summary of this and other models) and it uses optimal transport ideas to prove existence, and not to characterize the equilibria.

We want rather to concentrate on another, simple, class of models: suppose that a population ρ is distributed in a region Ω and that every individual wants to choose his own position in order to minimize the sum of two costs:

- a residential cost, which is an increasing function of the density of population at the place where he lives; the individuals living at x “pay” a function of the form $h(\rho(x))$, for $h : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ increasing; this takes into account the fact that where more people live, the price of land is higher (or that, for the same price, they have less space);
- an interaction cost, depending on the distances with all the other individuals; people living at x “pay” a cost of the form $\int W(x-y)\rho(y) dy$ where W is usually an increasing function of the distance (i.e. $W(z)$ depends increasingly on $|z|$).

This means that the cost that every individual pays depends on his location x and on the whole distribution ρ of everybody. We look for an equilibrium configuration, i.e. a density ρ such that, for every x_0 , there is no reason for people at x_0 to move to another location, since the function $f_\rho(x) := h(\rho(x)) + \int W(x-y)\rho(y) dy$ is minimal at x_0 , in the spirit of Nash equilibria.

Box 7.3. – Important notion – Nash equilibria

Definition - Consider a game where several players $i = 1, \dots, n$ must choose a strategy among a set of possibilities S_i and suppose that the pay-off of each player (i.e. how much he gains out of the game) depends on what everybody chooses, i.e. it is given by a function $p_i : S_1 \times \dots \times S_n \rightarrow \mathbb{R}$. We say that a configuration $(s_1, \dots, s_n)$ (where $s_i \in S_i$) is an equilibrium (a *Nash equilibrium*) if, for every i , the choice s_i optimizes $S_i \ni s \mapsto f_i(s_1, \dots, s_{i-1}, s, s_{i+1}, \dots, s_n)$ (i.e. s_i is optimal for player i under the assumption that the other players freeze their choice).

Note that Nash equilibria need not exist in all situations, but the strength of Nash approach was exactly to prove (via convex analysis arguments) that they always exist when

we consider the so-called *mixed strategies*. This means that we accept that every player instead of choosing an element $s_i \in S_i$, only chooses a probability on S_i and then randomly picks a strategy according to the law he has chosen. This allows to convexify the game in a way which is similar to what we do when we pass from Monge to Kantorovich problems.

This notion, first introduced by J. Nash in [243, 244] in the case of a finite number of players, can be easily extended to a continuum of players where each one is negligible compared to the others (*non-atomic games*). Considering for simplicity the case of identical players, we have a common space S of possible strategies and we look for a measure $\rho \in \mathcal{P}(S)$. This measure induces a payoff function $f_\rho : S \rightarrow \mathbb{R}$ and we want the following condition to be satisfied: there exists $C \in \mathbb{R}$ such that $f_\rho(x) = C$ ρ -a.e. and $f_\rho(x) \leq C$ everywhere (if the players want to maximize the payoff, otherwise, if it is a cost to be minimized, $f_\rho(x) \geq C$), i.e. f_ρ must be optimal ρ -a.e.

It is easy to see that the equilibrium condition, namely that

$$h(\rho(x)) + \int W(x-y)\rho(y)dy$$

must be minimal ρ -a.e. is indeed the first-order optimality condition for the minimization of

$$\rho \mapsto F(\rho) := \int H(\rho(x))dx + \frac{1}{2} \iint W(x-y)\rho(y)dy\rho(x)dx$$

among probabilities $\rho \in \mathcal{P}(\Omega)$, where $H' = h$ (just compute the first variation with respect to perturbation of the form $\rho_\varepsilon = (1-\varepsilon)\rho + \varepsilon\tilde{\rho}$). We can wonder whether this optimality condition, which is necessary for minimality, is also sufficient, and if all equilibria are minimizers. This would allow to say that we face a *potential game* (i.e. a game where equilibria can be found by minimizing a global energy, see [238]). Note by the way that the energy that we minimize is not the total cost for residents, which would be given by $\int f_\rho(x)\rho(x)dx$ (compare to the notion of price of anarchy in Remark 4.29). This last quantity differs from what we consider both in the $\frac{1}{2}$ factor in the interaction part, and in the fact that $H(t) \neq th(t)$ in general (as we already saw in Section 4.4.1 concerning traffic congestion).

The answer to the above question would be affirmative if the functional F was convex. Yet, the first part is convex as H is convex (as it is the antiderivative of an increasing function), but the interaction energy is not, as we saw in this chapter. On the other hand, if H satisfies McCann's assumptions for displacement convexity and W is convex (which is the case for the interaction costs $|x-y|$ or $|x-y|^2$, for instance), then F is displacement convex. An interesting observation in [61] is exactly the fact that displacement convexity is also enough, as usual convexity would do, in order to guarantee that equilibria are all minimizers of F . To prove this fact it is enough to compute the derivative of $F(\rho_\varepsilon)$ at $\varepsilon = 0$, when we take $\rho_\varepsilon = ((1-\varepsilon)\text{id} + \varepsilon T)_\# \rho$, there T is the optimal transport map between ρ and any other admissible $\tilde{\rho}$. By the way, it is not a problem to add to the equilibrium condition a geographical heterogeneity (i.e. a potential function), taking $f_\rho(x) = h(\rho(x)) + \int W(x-y)\rho(y)dy + V(x)$, and in this

case we add to $F(\rho)$ a term in $\int V(x)\rho(x) dx$. The convexity of V would be the good assumption to the perform the same analysis.

Since we are evoking McCann's condition on the convex function H , it is worthwhile to spend some words on its meaning in economic terms. To do that, we need to check how h and H depend on the density through the formation of the price of land. This comes from an easy consideration: suppose that at every point, the agents have a certain budget to be divided into land consumption and money consumption, and that they have a concave and increasing utility function U for land. This means they solve a problem of the form

$$\max\{U(L) + m : pL + m \leq 0\},$$

where p represents the price for land, L is the land consumption, m is the left-over of the money, and the budget constraint has been set to 0 for simplicity. The optimal land consumption will saturate the constraint, and be such that $U'(L) = p$. This allows to compute the optimal utility, which is $U(L) - U'(L)L$ and this gives a relation between L and utility. Yet, the land consumption is the reciprocal of the density, hence $L = \frac{1}{\rho}$, and the residential cost $h(\rho)$, which is a cost and not a utility, can be taken to be opposite as this utility. Hence, we take

$$h(\rho) = \frac{1}{\rho} U' \left(\frac{1}{\rho} \right) - U \left(\frac{1}{\rho} \right).$$

Since the function $t \mapsto \frac{1}{t} U' \left(\frac{1}{t} \right) - U \left(\frac{1}{t} \right)$ is the derivative of $-tU \left(\frac{1}{t} \right)$, the correct choice for H is indeed $H(t) = -tU \left(\frac{1}{t} \right)$. This makes the verification of McCann's condition for displacement convexity very easy, since $t \mapsto t^d H(t^{-d})$ becomes $t \mapsto -U(t^d)$ and we just need $t \mapsto U(t^d)$ to be concave and increasing. Monotonicity is guaranteed since U is monotone, while the fact that it stays concave when composed with the d -th power is indeed an extra assumption⁵.

We also stress that, in order to use displacement convexity, one needs a convex domain Ω and that [61] also presents examples of multiple equilibria when the convex domain is replaced by a one-dimensional circle (equilibria with multiple sub-cities are found, i.e. $\text{spt}(\rho)$ can be disconnected, and the multiplicity of the equilibria is not only due to rotational invariance, but equilibria with different shapes and different number of components are found). Curiously enough, the model described in [61] strongly recalls a model by R. McCann for rotating stars, described in [232]. In such a paper, the author introduces a notion of W_∞ -local minimizers which can be translated into a notion of local equilibrium: each agent only compares his cost to that of nearby agents; should the support of ρ be disconnected (which is the case in [61] and [232]), then the constant value of their utility could change from one connected component to the others. Note the early use of the W_∞ distance, and the fact that this is the only distance allowing for this kind of local phenomena. Indeed, local minimality for integral distances such as W_p , if translated via optimality conditions into the expression of an equilibrium, would not provide local minimality for each agent, but automatically global minimality. This

⁵Thanks to an observation by R. McCann himself, this corresponds to the fact that the utility is a concave function not only of land "volume" but also of land "linear size", which seems reasonable since it has to be compared to "linear" quantities such as distances in the interaction term.

can be observed in the example of the circle in [61]: in this example many equilibria exist, and each agent is globally satisfied, not only locally, but most of them are not global minimizers of the global energy.

An even more interesting situation is presented in [58], in the framework of the so-called *Cournot-Nash* equilibria. This name is used to denote nonatomic games with agents which are not indistinguishable. The situation is similar to the one described above, but with an extra variable, which makes the distinction between agents: suppose that a population of agents μ distributed over a space Z must choose, each, a location $x \in \Omega$, and that the payoff of each agent z is given by $c(z, x) + f_\rho(x)$, where ρ is the distribution of their choices on Ω . More precisely, we look for a measure $\gamma \in \mathcal{P}(Z \times \Omega)$ such that $(\pi_Z)_\# \gamma = \mu$, and we denote $(\pi_\Omega)_\# \gamma$ by ρ . The goal is to find γ such that no agent z want to change his choice, i.e. there is a function $\varphi : Z \rightarrow \mathbb{R}$ with $c(z, x) + f_\rho(x) \geq \varphi(z)$ for all (z, x) and $c(z, x) + f_\rho(x) = \varphi(z)$ γ -a.e. Note that the variable z represents the type of the agent, which cannot be modified by the agent itself: this is why the minimality of $c(z, x) + f_\rho(x)$ is tested only on x , and gives a minimal value depending on z .

It is not difficult to prove the following facts (see [58]): a measure γ is an equilibrium if and only if it is optimal for the cost c in $\Pi(\mu, \nu)$, the pair $(\varphi, -f_\rho)$ is a pair of Kantorovich potentials for the same transport problems, and the c -transform φ^c satisfies $\varphi^c(x) + f_\rho(x) \geq 0$ for all x and $\varphi^c(x) + f_\rho(x) = 0$ ρ -a.e. This leads to a condition on ρ only: find ρ such that $\psi + f_\rho$ is minimal ρ -a.e., where $\psi = \varphi^c$ is the Kantorovich potential in the transport from ρ to the given measure μ , for the cost c .

The same considerations of this chapter allow to view this as the optimality condition for the minimization of $\rho \mapsto \mathcal{T}_c(\mu, \rho) + F(\rho)$ and the same questions about displacement convexity hold. As we saw in Section 7.3.3, the functional $\mathcal{T}_c(\mu, \cdot)$ is not in general displacement convex, and one needs to use convexity on generalized geodesics. But the computations and the result are similar, as one can see in [58].

Finally, it is interesting to see that the same kind of models would appear if, instead of defining an equilibrium on agent types z who must choose x , one directly sets an equilibrium problem on x : agents have to choose where to locate, facing both a residential cost and an interaction cost, but also need to access a commodity z , distributed according to a distribution μ , and their access cost is $c(x, z)$. Since the distribution of the commodity is fixed, the agents cannot simply choose the z that they prefer (i.e. minimizing $z \mapsto c(x, z)$): a matching problem will appear, and a price $\varphi(z)$ for each commodity. At equilibrium, according to the theory that we saw in Section 1.7.3, the commodity cost for the agents will be given by $\varphi(z) + c(x, z) = \varphi^c(y)$. This means that the agents add to their cost a new term, given by the Kantorovich potential φ^c , exactly as above.

Chapter 8

Gradient Flows

In this chapter we present one of the most spectacular application of optimal transport and Wasserstein distances to PDEs. We will see that several evolution PDEs can be interpreted as a steepest descent movement in the space $\mathbb{W}_2$. As we already developed in the previous chapters many of the technical tools that we need, a large part of this chapter will be devoted to an informal description of the general framework.

Hence, no need to let the reader wait any more, let us immediately enter into the subject!

8.1 Gradient flows in $\mathbb{R}^d$ and in metric spaces

First of all, let us present what a gradient flow is in the simplest situation. Suppose you have a function $F : \mathbb{R}^d \rightarrow \mathbb{R}$ and a point $x_0 \in \mathbb{R}^d$. A gradient flow is an evolution stemming from x_0 and always moving in the direction where F decreases the most, thus “gradually minimizing” F , starting from x_0 . More precisely, it is just the solution of the Cauchy problem

$$\begin{cases} x'(t) = -\nabla F(x(t)) & \text{for } t > 0, \\ x(0) = x_0. \end{cases}$$

This is a standard Cauchy problem which has a unique solution if ∇F is Lipschitz continuous, i.e. if $F \in C^{1,1}$. We will see that existence and uniqueness can also hold without this strong assumption, thanks to the variational structure of the equation.

A first interesting property is the following, concerning uniqueness and estimates.

Proposition 8.1. *Suppose that F is convex and let x_1 and x_2 be two solutions of $x'(t) = -\nabla F(x(t))$ (if F is not differentiable we can consider $x'(t) \in \partial F(x(t))$). Then we have $|x_1(t) - x_2(t)| \leq |x_1(0) - x_2(0)|$ for every t . In particular this gives uniqueness of the solution of the Cauchy problem.*

Proof. Let us consider $g(t) = \frac{1}{2}|x_1(t) - x_2(t)|^2$ and differentiate it. We have

$$g'(t) = (x_1(t) - x_2(t)) \cdot (x_1'(t) - x_2'(t)) = -(x_1(t) - x_2(t)) \cdot (\nabla F(x_1(t)) - \nabla F(x_2(t))).$$